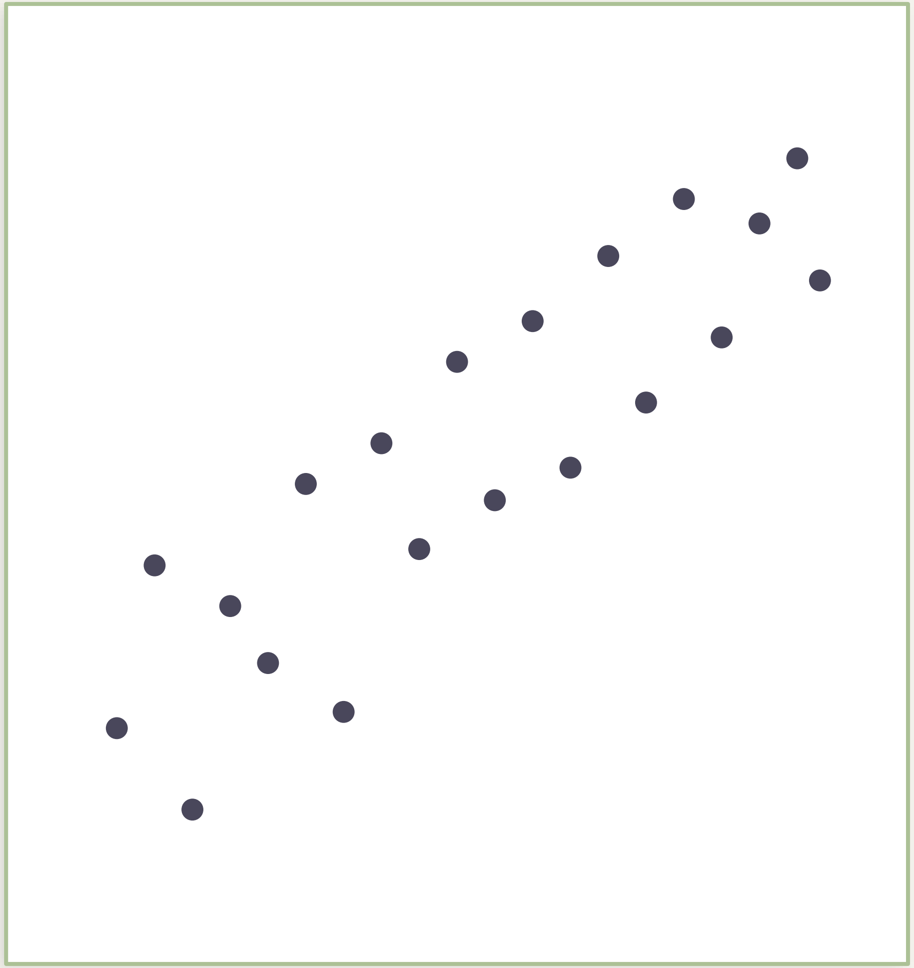
StatQuest

how well do you really know your stats?

HSLU IIM STCSA · Final Project · Kai Steinke & Mauro Zappa · 11.05.2026

What's the correlation?

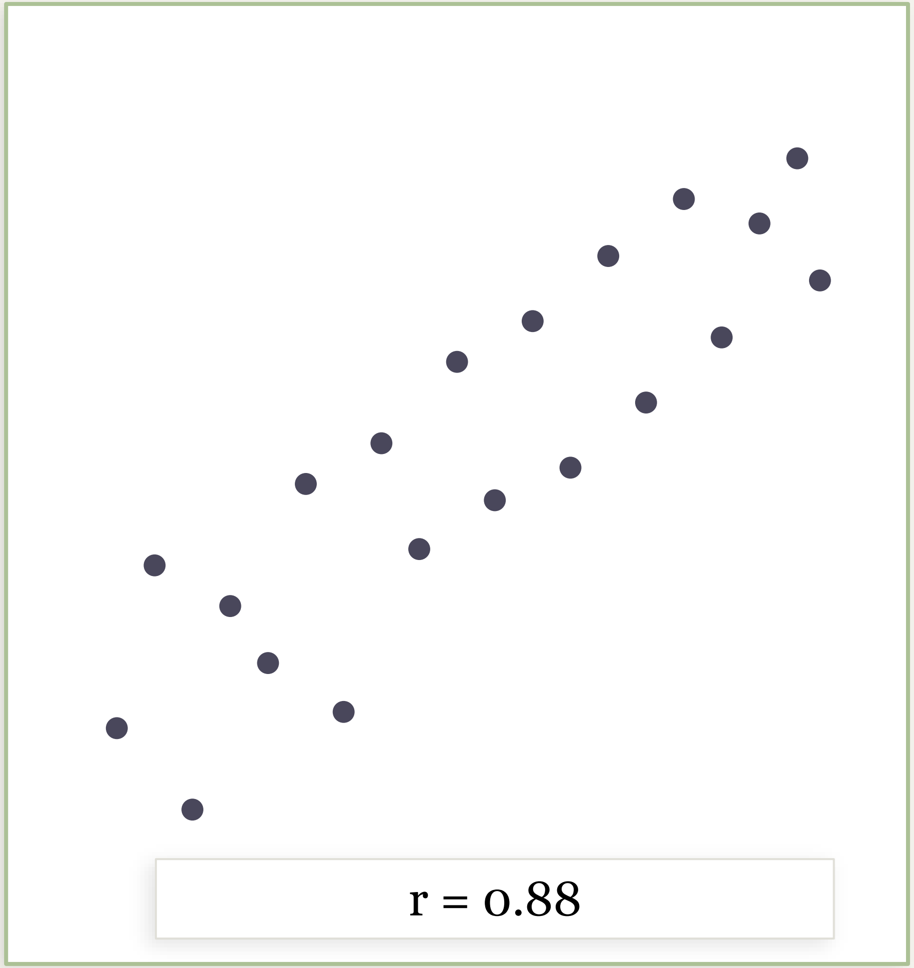
shout a number.



That's the problem.

Statistics is something we calculate, not something we feel.

We wanted to change that.



We built three mini-games.

Guess the Correlation

Use a slider to estimate r from a live scatter plot. 5 rounds.

200 pts / round

Name That Distribution

Identify whether a histogram is normal, skewed, bimodal, or uniform.

100 pts / round

Spot the Outlier

Click the data point that doesn't belong. Z-score reveal after.

150 pts / round

Nothing is hardcoded.

CORRELATION

Covariance matrix → mathematically guaranteed r, verified with scipy.stats.pearsonr()

DISTRIBUTION

scipy.stats.skew() confirms the shape before every round — no accidental normals

OUTLIER

Outliers injected at 2.5 – 4σ so difficulty is precise, not just visual *demo.*

What we learned.

PYTHON

Session state is harder than it looks.

STATISTICS

Small samples lie.

Statistics is a lot more intuitive when you play with the data.

Thank you for your attention.

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